



Syllabus

CS 300 - Data Structures

Fall 2025-2026

Faculty	Faculty of Engineering and Natural Sciences			
Semester	Fall 2025-2026			
Course Code	CS 300			
Course Name	Data Structures			
Time/Place	Time	Week Day	Place	Date
	12:40-14:30	Tue	UC-G030	Sep 29, 2025-Jan 3, 2026
	14:40-15:30	Fri	SBS-G071	Sep 29, 2025-Jan 3, 2026
Level of Course	Undergraduate			
Credits	SU Credit:3, ECTS:6, Basic:1, Engineering:5			
Prerequisites	CS 204			
Corequisites	CS 300R			
Course Type	Lecture			
Instructors	Name	Email		
	Cemal Yılmaz	cyilmaz@sabanciuniv.edu		
Course Description	This course covers some of the fundamental data structures as well as the algorithms operating on them. Topics included are: Introduction to theoretical aspects of computing: modeling algorithms and their run times, computational complexity. Linear data structures (lists, stacks, queues), trees (binary search trees, AVL trees, tree traversals), hash tables, priority queues, disjoint sets, sorting algorithms, graphs and their representations, graph algorithms (depth first and breadth first search, single source shortest path algorithms), sorting algorithms, algorithmic paradigms (divide and conquer, greedy, dynamic programming).			
Learning Outcomes	1. describe common data structures and associated algorithms			
	2. contrast the advantages and disadvantages of these data structures			
	3. apply these data structures to develop algorithmic solutions to practical problems			
	4. design new data structures and associated algorithms			
	5. analyze the computational complexity of simple algorithms and programs			
Course Objective	To introduce fundamental techniques of algorithm analysis To introduce common data structures, their properties and implementation To apply C++ programming skills to implement solutions using common data structures to practical problems			
Sustainable Development Goals (SDGs) Related to This Course	-			
Resources	Data Structures & Algorithm Analysis in C++, (2nd, 3rd, or 4th edition), Mark Allen Weiss, Pearson, 2014, ISBN 0-273-76938-3 (4th edition).			
Link to LMS/SUCourse	https://sucourse.sabanciuniv.edu/plus/course/view.php?id=12134			

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